



College of Engineering  
UNIVERSITY OF WISCONSIN-MADISON

# ECE/CS 552 Fall 2024 Syllabus

## Component 1: Key Course Offering Information

### General Identifying Information

**Institution Name:** University of Wisconsin–Madison

**Course Subject, Number and Title:** Introduction to Computer Architecture, ECE/CS 552

**Credits:** 3

#### Course Designations and Attributes:

Breadth - Physical Sci. Counts toward the Natural Sci req

Level - Advanced

L&S Credit - Counts as Liberal Arts and Science credit in L&S

**Course Description:** The design of computer systems and components. Processor design, instruction set design, and addressing; control structures and microprogramming; memory management, caches, and memory hierarchies; and interrupts and I/O structures.

**Requisites:** ([E C E/COMP SCI 352](#) and [E C E/COMP SCI 354](#)) or graduate/professional standing. Recommended: ECE 551 or knowledge of Verilog

**Meeting Time and Location:** Lecture: Tuesday/Thursday 9:30 AM – 10:45 AM 2540 Engineering Hall

**Instructional Modality:** Blended. Students are expected to watch online video lectures ahead of time and complete in-class exercises during designated class hours.

**Instructor Contact Info:** Mr. Boone Severson; Office hours 8:30 AM – 9:25 AM 4613 Engineering Hall; [bseverson3@wisc.edu](mailto:bseverson3@wisc.edu); available via Zoom mornings/evenings by appointment

**Teaching Assistant Contact Info (if applicable):** Shreya Malkurthi, Wednesday 3:30 PM – 4:30 PM 209 Animal Science Building, [shreya.malkurthi@wisc.edu](mailto:shreya.malkurthi@wisc.edu)

### Course Learning Outcomes

The student will be able to:

- Use standard performance metrics to compare performance of different digital systems.
- Design a pipelined data path for a RISC (reduced instruction set computer) instruction set and be familiar with concepts of data dependence, pipelined hazards and out of order execution.
- Design basic data and control cache subsystems and understand basic memory organization.
- Design a pipelined RISC micro-processor system with data cache using computer aided design tool and validate the correctness of the design using logic simulation.

At the time of completion, ECE/CS 552 students will have attained:

- An ability to apply knowledge of mathematics, science, and engineering
- An ability to design and conduct experiments, as well as to analyze and interpret data

- An ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
- An ability to identify, formulate, and solve engineering problems
- An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.

### How Credit Hours are Met by the Course

This course follows the Traditional Carnegie Definition. The class meets for two 75-minute class periods each week over the semester and carries the expectation that students will work on course learning activities (reading, writing, problem sets, studying, etc) for about 3 hours out of classroom for every class period. The syllabus includes more information about meeting times and expectations for student work.

## Component 2: Instructor-to-Student Communication

### Course Website and Digital Instructional Tools

- **Course Website:** <https://canvas.wisc.edu/courses/430867>
  - All video lectures, course material and announcements will be posted on the University's learning management system, [Canvas](#). It is strongly suggested that students explore and become familiar not only with Canvas' site navigation but with content and resources available for the course.
  - Students are expected to watch online video lectures prior to class.
  - Class time will be spent on in-class exercises. Students are required to bring their own computing device (laptop) in order to complete the exercises. Attendance is mandatory.
- By-appointment office hours will use [Zoom](#) or [MS Teams](#)

### Discussion and/or Laboratory Sessions

- The **Optional** Discussion section will be Wednesday 3:30 PM – 4:30 PM 209 Animal Science Building.
- Laboratory Sessions are not applicable to this course.

### Required Textbook, Software and Other Course Materials

- Computer Organization and Design, Sixth Edition, David Patterson and John Hennessy (MIPS Edition)
- **Canvas > Course Project > Verilog Resources for Project** page discusses the use of software used to simulate Verilog RTL.

### Homework and Other Assignments

- There will be 11 weekly homework assignments.
- The project is to implement a working pipelined processor that implements a custom instruction set. The project is to be completed in groups of three students; no individual projects will be allowed. Successful completion of the project includes a demonstration and a written report that documents your design.
- All homework & project phases are submitted via Canvas

### Exams, Quizzes, Papers and Other Major Graded Work

- The midterm will be held in-class 9:30-10:45 AM on Thursday October 24, 2024
- The final exam will be held 12:25-2:25 PM on Tuesday, December 17, 2024
- Weekly quizzes will test your knowledge of online lectures, which you are expected to watch beforehand.

- Exercises will be completed in-class and submitted for credit. Attendance is required. The lowest three exercise scores will be dropped for the final grade.
- List the summary period and the expectations associated with it. View the [Summary Period for Academic Semester Policy](#).
- Include relevant details such as dates, if the exam or quiz is cumulative, open-book or open-note, whether access to electronic devices is allowed, etc.
- Note if exams or quizzes will be proctored (see the “General Guidelines for Exam Proctoring” section below).
- Explain policies for make-up and/or late work.

## Course Schedule/Calendar

The [Weekly Outline page](#) in Canvas contains the course schedule.

## Grading

- 10% In-Class Exercises (attendance required)
- 10% Homework
- 10% Weekly Quizzes
- 20% Midterm
- 20% Project
- 30% Final Exam
- Letter Grade Curved

## Component 3: Academic Policies and Statements

Institutional academic policies and statements are reviewed and updated annually, as needed. They currently include:

- Academic Calendar and Religious Observances
  - View the full [academic calendar](#) in addition to information about religious and election day observances. Students are responsible for notifying instructors within the first two weeks of classes about any need for flexibility due to [religious observances](#).
  - Establishment of the academic calendar for the University of Wisconsin–Madison falls within the authority of the faculty as set forth in [Faculty Policies and Procedures](#). Construction of the academic calendar is subject to various rules and laws prescribed by the Board of Regents, the Faculty Senate, State of Wisconsin and the federal government. Find [additional dates and deadlines for students](#) on the Office of the Registrar website.
- Academic Integrity
  - By virtue of enrollment, each student agrees to uphold the high academic standards of the University of Wisconsin–Madison; academic misconduct is behavior that negatively impacts the integrity of the institution. Cheating, fabrication, plagiarism, unauthorized collaboration and helping others commit these previously listed acts are examples of misconduct which may result in disciplinary action. Examples of [disciplinary sanctions](#) include, but are not limited to, failure on the assignment/course, written reprimand, disciplinary probation, suspension or expulsion.
- Accommodations for Students with Disabilities
  - The University of Wisconsin–Madison supports the right of all enrolled students to a full and equal educational opportunity. The Americans with Disabilities Act (ADA), Wisconsin State Statute

(36.12) and UW–Madison policy ([UW-855](#)) require the university to provide reasonable accommodations to students with disabilities to access and participate in its academic programs and educational services. Faculty and students share responsibility in the accommodation process. Students are expected to inform faculty of their need for instructional accommodations during the beginning of the semester, or as soon as possible after being approved for accommodations. Faculty will work either directly with the student or in coordination with the [McBurney Disability Resource Center](#) to provide reasonable instructional and course-related accommodations. Disability information, including instructional accommodations as part of a student’s educational record, is confidential and protected under FERPA.

- Course Evaluations
  - Students at the University of Wisconsin–Madison have the opportunity to evaluate their learning experiences and the courses they are enrolled in through course evaluations. Many instructors use a [digital course evaluation tool](#) to collect feedback from students. Students typically receive notifications two weeks prior to the end of the semester requesting that they complete course evaluations. Student participation is an integral component of course development, and confidential feedback is important. UW–Madison strongly encourages student participation in course evaluations.
- Diversity and Inclusion
  - [Diversity](#) is a source of strength, creativity, and innovation for the University of Wisconsin–Madison. We value the contributions of each person and respect the profound ways their identity, culture, background, experience, status, abilities, and opinion enrich the university community. We commit ourselves to the pursuit of excellence in teaching, research, outreach, and diversity as inextricably linked goals. UW–Madison fulfills its public mission by creating a welcoming and inclusive community for people from every background – people who as students, faculty, and staff serve Wisconsin and the world.
- Mental Health and Well-Being
  - Students often experience stressors that can impact both their academic experience and personal well-being. These may include mental health concerns, substance misuse, sexual or relationship violence, family circumstances, campus climate, financial matters, among others.
  - UW–Madison students are encouraged to learn about and utilize the university’s mental health services and/or other resources as needed. Student can visit [uhs.wisc.edu](#) or call University Health Services at (608) 265-5600 to learn more.
- Privacy of Student Records and the Use of Audio Recorded Lectures
  - Lecture materials and recordings for this course are protected intellectual property at UW–Madison. Students enrolled in this course may use the materials and recordings for their personal use related to participation in the course. Students may also take notes solely for their personal use. If a lecture is not already recorded, students are not authorized to record lectures without permission unless they are considered by the university to be a qualified student with a disability who has an approved accommodation that includes recording. [Regent Policy Document 4-1] Students may not copy or have lecture materials and recordings outside of class, including posting on internet sites or selling to commercial entities, with the exception of sharing copies of personal notes as a notetaker through the McBurney Disability Resource Center. Students are otherwise prohibited from providing or selling their personal notes to anyone else or being paid for taking notes by any person or commercial firm without the instructor’s express written permission.

Unauthorized use of these copyrighted lecture materials and recordings constitutes copyright infringement and may be addressed under the university's policies, UWS Chapters 14 and 17, governing student academic and non-academic misconduct. View [more information about FERPA](#).

- Students' Rules, Rights and Responsibilities
  - View more information about [student rules, rights and responsibilities](#) such as student privacy rights, sharing of academic record information, academic integrity and grievances.
- Teaching and Learning Data Transparency
  - The privacy and security of faculty, staff and students' personal information is a top priority for UW–Madison. The university carefully reviews and vets all campus-supported digital tools used for teaching and learning, including those that support [data empowered educational practices](#) and proctoring. View more information about [teaching and learning data transparency](#) at UW–Madison.